

Eagle OTR Ops Suite - Architecture Documentation

Project Structure

```

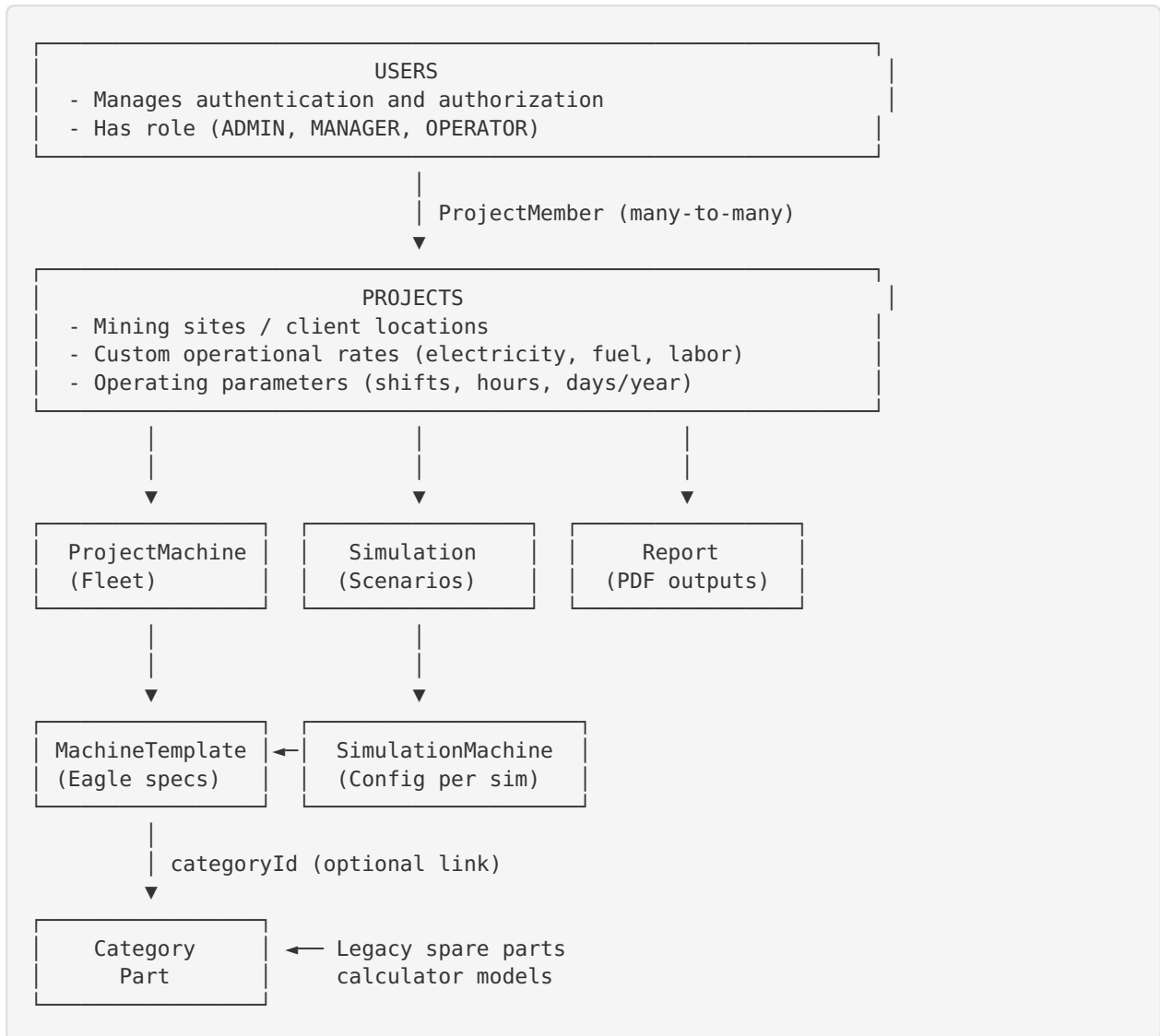
eagle_spare_parts_calculator/
├── nextjs_space/
│   ├── app/
│   │   ├── (auth)/
│   │   │   ├── login/
│   │   │   └── signup/
│   │   ├── (dashboard)/
│   │   │   ├── projects/
│   │   │   │   ├── page.tsx
│   │   │   │   ├── new/
│   │   │   │   └── [id]/
│   │   │   │       ├── page.tsx
│   │   │   │       ├── fleet/
│   │   │   │       ├── simulations/
│   │   │   │       └── reports/
│   │   │   ├── simulations/
│   │   │   │   ├── page.tsx
│   │   │   │   ├── new/
│   │   │   │   └── [id]/
│   │   │   │       ├── page.tsx
│   │   │   │       └── edit/
│   │   │   ├── reports/
│   │   │   │   ├── page.tsx
│   │   │   │   └── [id]/
│   │   │   └── settings/
│   │   ├── calculator/
│   │   └── history/
│   ├── api/
│   │   ├── auth/
│   │   ├── projects/
│   │   │   ├── route.ts
│   │   │   └── [id]/
│   │   │       ├── route.ts
│   │   │       ├── fleet/
│   │   │       └── members/
│   │   ├── machines/
│   │   │   └── route.ts
│   │   ├── simulations/
│   │   │   ├── route.ts
│   │   │   ├── calculate/
│   │   │   │   └── [id]/
│   │   │   │       └── route.ts
│   │   ├── reports/
│   │   │   ├── route.ts
│   │   │   ├── generate/
│   │   │   │   └── [id]/
│   │   │   │       └── route.ts
│   │   ├── parts/
│   │   ├── scenarios/
│   │   ├── production/
│   │   └── history/
└── # Next.js App Router
    ├── # Auth group (login/signup)
    ├── # Protected dashboard routes
    ├── # Project management
    ├── # Projects list
    ├── # Create project
    ├── # Project details
    ├── # Project overview
    ├── # Fleet management
    ├── # Simulations for this project
    ├── # Reports for this project
    ├── # All simulations
    ├── # Simulations list
    ├── # Create simulation
    ├── # Simulation details
    ├── # Results view
    ├── # Edit simulation
    ├── # Report management
    ├── # Reports list
    ├── # Report viewer
    ├── # User/org settings
    ├── # Legacy spare parts calculator
    ├── # Calculation history
    ├── # API Routes
    ├── # Authentication endpoints
    ├── # Project CRUD
    ├── # GET all, POST new
    ├── # GET, PUT, DELETE
    ├── # Fleet management
    ├── # Project members
    ├── # Machine templates
    ├── # GET all templates
    ├── # Simulation CRUD
    ├── # GET all, POST new
    ├── # Run calculation
    ├── # GET, PUT, DELETE
    ├── # Report generation
    ├── # GET all, POST new
    ├── # Generate PDF
    ├── # GET, DELETE
    ├── # Legacy parts API
    ├── # Legacy scenarios API
    ├── # Production calculator API
    └── # Calculation history API

```

layout.tsx	# Root layout
page.tsx	# Landing page
globals.css	# Global styles
components/	
ui/	# Shadcn/UI components
button.tsx	
card.tsx	
dialog.tsx	
header.tsx	
...	# Other UI primitives
projects/	# Project-specific components
project-card.tsx	
project-form.tsx	
project-rates-form.tsx	
project-members.tsx	
fleet/	# Fleet management components
machine-card.tsx	
fleet-overview.tsx	
add-machine-modal.tsx	
simulations/	# Simulation components
simulation-builder.tsx	
machine-selector.tsx	
results-dashboard.tsx	
bottleneck-chart.tsx	
cost-breakdown.tsx	
comparison-view.tsx	
reports/	# Report components
report-builder.tsx	
report-preview.tsx	
pdf-template.tsx	
charts/	# Visualization components
production-gauge.tsx	
cost-pie-chart.tsx	
utilization-bars.tsx	
trend-line.tsx	
calculator/	# Legacy calculator components
parts-table.tsx	
results-panel.tsx	
production-calculator.tsx	
layout/	# Layout components
sidebar.tsx	
dashboard-header.tsx	
mobile-nav.tsx	
providers.tsx	# Context providers
lib/	
db.ts	# Prisma client
auth-options.ts	# NextAuth config
utils.ts	# Utility functions
types.ts	# TypeScript types
calculations/	# Business logic
production.ts	# Production calculations
costs.ts	# Cost calculations (OPEX)

		 bottleneck.ts	 Theory of Constraints
		 revenue.ts	 Revenue/ROI projections
		constants/	 Configuration constants
		 machines.ts	 Eagle machine specs
		 tire-weights.ts	 Tire weight categories
		 defaults.ts	 Default values
		pdf/	 PDF generation
		templates/	
		 generator.ts	
		hooks/	 Custom React hooks
		 use-project.ts	
		 use-simulation.ts	
		 use-calculations.ts	
		 use-toast.ts	
		prisma/	
		 schema.prisma	 Database schema
		 migrations/	 DB migrations
		scripts/	
		 seed.ts	 Database seeding
		 seed-machines.ts	 Seed machine templates
		public/	
		 favicon.svg	
		images/	
		 machines/	 Machine images
		config files...	

Database Schema (ERD)



Key Entities Explained

1. Project (Mining Site)

Each project represents a client location with unique operational costs:

Field	Type	Description
name	String	Project name (e.g., "Minera Escondida - Antofagasta")
location	String	Geographic location
clientName	String	Client company name
electricityRate	Float	\$/kWh (varies by region)
fuelRate	Float	\$/gallon for diesel equipment
laborRate	Float	\$/hour for operators
shiftsPerDay	Int	1, 2, or 3 shift operations
operatingDaysYear	Int	Typically 250-350 days
avgTireWeightKg	Float	Average tire weight at this site

2. MachineTemplate (Eagle Specs)

Official Eagle machine specifications from technical documentation:

Machine	Throughput	Power	Operators
OTR Debeader	6-8 tires/hr	75 kW	1
Punch Cutter II	4-6 tires/hr	55 kW	1
Titan II	4-5 tires/hr	90 kW	1

3. ProjectMachine (Fleet)

Actual equipment deployed at each site:

```
{
  projectId: "proj_123",
  templateId: 1,           // References MachineTemplate
  quantity: 2,             // Two machines of this type
  serialNumbers: ["SN001", "SN002"],
  status: "OPERATIONAL",
  customThroughput: null   // Override if different from spec
}
```

4. Simulation (Cost/Production Analysis)

Saved calculations for comparison:

Inputs:

- Selected machines and quantities

- Operating parameters
- Rate type (typical vs. maximum)

Outputs:

- Production metrics (tires/shift, tons/year)
- Cost breakdown (labor, energy, maintenance)
- Bottleneck identification
- ROI projections

5. Report (PDF Generation)

Professional reports for client presentation:

Type	Contents
PRODUCTION_ANALYSIS	Throughput, bottlenecks, utilization
COST_BREAKDOWN	OPEX details, cost per tire/ton
ROI_PROJECTION	Revenue potential, payback period
EXECUTIVE_SUMMARY	High-level overview for decision makers

 Data Flow

1. User creates PROJECT with site-specific rates

2. User adds FLEET (machines from templates)

3. User creates SIMULATION

- Selects machines from fleet
- Sets operating parameters
- System calculates:
 - Bottleneck (Theory of Constraints)
 - Production rates
 - Annual costs (OPEX)
 - Revenue potential

4. User generates REPORT

- Selects simulation data
- Customizes sections
- Exports PDF for client

Technology Stack

Layer	Technology
Frontend	Next.js 14 (App Router), React 18, TypeScript
Styling	Tailwind CSS, Shadcn/UI
State	React Query, Zustand
Database	PostgreSQL (Abacus.AI hosted)
ORM	Prisma
Auth	NextAuth.js (Credentials)
Charts	Recharts
PDF	Client-side print / html2pdf

Next Steps

1. **Seed machine templates** - Add Eagle machine specs to DB
2. **Build Projects module** - CRUD for projects/sites
3. **Build Fleet module** - Machine inventory per project
4. **Build Simulations module** - Advanced cost calculator
5. **Build Reports module** - PDF generation
6. **Add Dashboard** - Overview with key metrics